RAJALAKSHMI ENGINEERING COLLEGE

Priyadharshini K 250701555-2025-CSE

P2

Started on	Sunday, 30 August 2026, 2:21 PM
State	Finished
Completed on	Sunday, 30 August 2026, 2:23 PM
Time taken	2 mins 34 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two arrays array_One[] and array_Two[] of same size N. We need to first rearrange the arrays such that the sum of the product of pairs(1 element from each) is minimum. That is SUM (A[i] * B[i]) for all i is minimum.

For example:

Input	Result
3 1 2 3 4 5 6	28

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int ascending(const void *a, const void *b) {
    int x = *(const int *)a;
    int y = *(const int *)b;
    return x - y;
}

int descending(const void *a, const void *b) {
    int x = *(const int *)a;
    int y = *(const int *)b;
    return y - x;
}

int main() {
    int n;
    scanf("%d", &n);

    int a[n], b[n];

    // Read first array
    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    // Read second array
    for (int i = 0; i < n; i++) {
        scanf("%d", &b[i]);
    }

    // Sort first array ascending
    qsort(a, n, sizeof(int), ascending);

    // Sort second array descending
    qsort(b, n, sizeof(int), descending);

    long long sum = 0;

    for (int i = 0; i < n; i++) {
        sum += (long long)a[i] * b[i];
    }

    printf("%lld\n", sum);

    return 0;
}

```

	Input	Expected	Got	
✔	3 1 2 3 4 5 6	28	28	✔
✔	4 7 5 1 2 1 3 4 1	22	22	✔
✔	5 20 10 30 10 40 8 9 4 3 10	590	590	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course